

A Tangible Mixed Reality System for 3D Modeling

Mohd Zubair*

Creative Interfaces Lab, IIIT Delhi

Vishal Bharti†

Creative Interfaces Lab, IIIT Delhi

Anmol Srivastava‡

Creative Interfaces Lab, IIIT Delhi

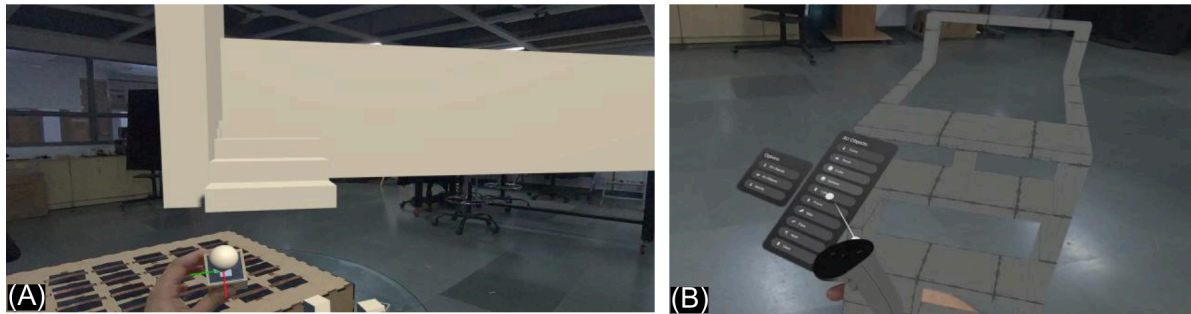


Figure 1: (A) Tangible token-based interface, where physical tokens with overlaid 3D previews are used to create and modify models on the BoxCraft surface in MR. (B) Controller-based interface, where a Meta Quest 3 controller is used for 3D modeling in MR.

ABSTRACT

3D modeling using traditional computer-aided design (CAD) tools often poses challenges, particularly to novice users. Several prior works have explored virtual reality-based CAD systems. However, most face major problems such as loss of accuracy. To address these challenges, we present BoxCraft a tangible mixed-reality (MR) system for intuitive 3D modeling with two modalities. BoxCraft allows users to create and edit 3D models in an MR space by using physical tokens enhanced with visual overlays for immediate spatial feedback. We conducted a participatory design workshop ($n = 6$) and an exploratory user study ($n = 7$) to gather initial qualitative feedback on interaction design and user experience. Preliminary results show that, although tangible token-based interaction requires a longer initial learning phase, participants, especially beginners, preferred it over controller-based interaction and traditional CAD tools for its intuitiveness and physical affordances. This research demo presents the BoxCraft system, its interaction techniques, and initial user feedback, pointing out the potential of tangible MR interfaces in making 3D modeling more accessible.

Index terms: Mixed Reality, Tangible User Interface, 3D Modeling, CAD Modeling, Tangible Mixed Reality.

1 INTRODUCTION

3D modeling using traditional CAD software remains challenging for many users, particularly beginners, due to complex interfaces and steep learning curves [1]. Although VR-CAD modeling tools are available, these tools are not regularly used to create CAD designs due to their low precision and high hardware cost [2]. However, there is limited exploration of tangible interaction in mixed reality systems for novice-friendly CAD modeling. To solve

this problem, we developed BoxCraft, a tangible mixed reality system. Initial design exploration suggested that tangible interaction could support understanding of complex 3D designs, particularly for beginners. Through the participatory design workshop, users reported that they were able to make 3D models more effectively due to a simplified interface. The system supported two interaction modes for 3D modeling, which were controller-based and tangible token-based. Controller-based interaction was performed using a handheld controller, while tangible interaction was achieved using physical tokens. Depending on the user's comfort, either interaction mode could be selected.

2 SYSTEM OVERVIEW

BoxCraft is a system used to create 3D-CAD designs and visualize them in MR. On the surface of BoxCraft, an FSR (*Force Sensitive Resistor*) sensor was integrated to detect pressure applied by a tangible token. A Meta Quest 3 headset was connected to visualize the 3D model in the MR environment. A computer system or laptop was required to connect BoxCraft to Meta Quest 3. When a user places a token on the BoxCraft surface, the applied pressure is sensed and the ArUco marker is detected using the Meta Quest 3 camera. After this, the corresponding 3D object or operation (e.g., translation, modification) is performed. Every token had an ArUco marker attached on top of it, whether it was a 3D model token or an operational token. A 3D model was overlaid on the tangible token so that before even making the model in MR space a user could get an idea of how the shape appeared. Therefore, the chances of error were reduced during the design process (see Figure 2).

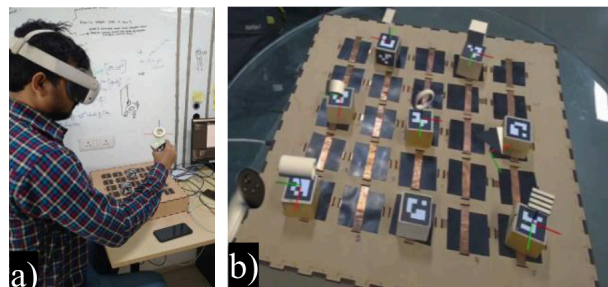


Figure 2: a) User interacting with a tangible token with a 3D model overlaid on it, b) BoxCraft system with a token on its surface.

* e-mail: zubairi@iiitd.ac.in

† e-mail: vishal20264@iiitd.ac.in

‡ e-mail: anmol@iiitd.ac.in

2.1 Token

The BoxCraft system included two types of tokens:

1. **3D model token:** These tokens represent a 3D model, which was virtually overlaid on the token surface.
2. **Operational token:** These tokens helped users perform editing operations on a 3D model (e.g., extrusion, joining, undo) and used visual icons to represent their functions.

Initially, a rectangular wooden cube with an ArUco marker attached to one side was used. Later, multiple sides of the wooden cube were covered with ArUco markers. Feedback was received from a participatory design workshop and an initial user study. Based on this feedback, the token design was changed from a rectangular wooden cube to a 3D-printed token (see Figure 3). These 3D-printed tokens have similar shapes to their corresponding virtual counterparts. When a user placed a physical token on the BoxCraft surface, the corresponding virtual shape appeared in the MR environment. A virtual preview of the shape was overlaid on each model token to support user understanding. Magnets were embedded in the tokens to ensure stable attachment to the BoxCraft surface. The system included model tokens for common 3D primitives (e.g., cube, sphere, cylinder) and operational tokens for transformation and management actions.

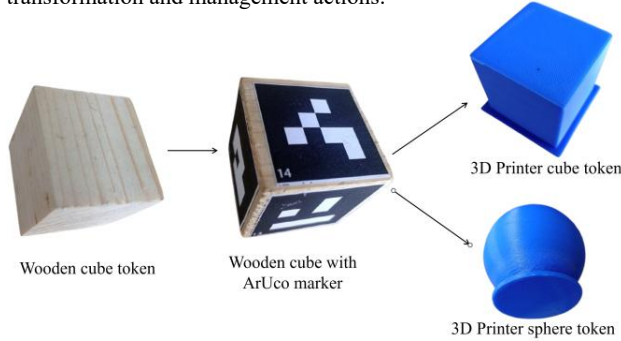


Figure 3: Evolution of tangible tokens from wooden and ArUco-marked tokens to 3D-printed shape tokens.

1 INTERACTION TECHNIQUES

To create a 3D shape, users had two interaction options:

1. **Controller-Based Interaction:** These interactions were performed using the Meta Quest 3 controller (Figure 4).

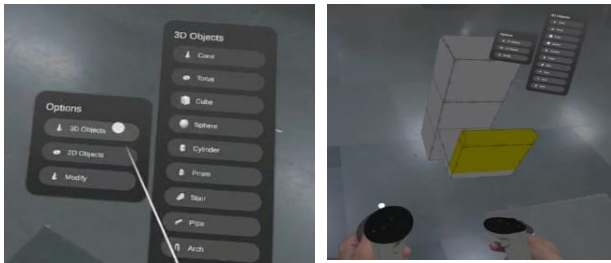


Figure 4: Controller-based interface for selecting modeling operations using the Meta Quest 3 controller.

2. **Tangible Token-Based Interaction:** Tangible token-based interactions were supported by placing physical tokens on the BoxCraft surface (see Figure 5), with supported gestures summarized in Table 1.

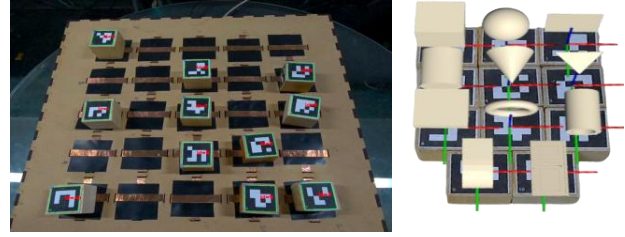


Figure 5: Tangible token-based interaction on the BoxCraft surface with an overlaid 3D model.

Tangible token-based interaction is more suitable for early-stage modeling and novice users because it supports spatial understanding through physical manipulation.

Table 1: Interaction design space supported by tangible tokens.

Category	Operations	Interaction design space
Translation	X, Y, or Z-Axis Movement	Select object, apply Translate marker and rotate secondary marker to define axis displacement.
Rotation	Clockwise/Counter-Clockwise	Select model, apply Rotation marker and rotate secondary marker in the desired direction.
Scaling	Uniform Scaling	Select object, apply variable pressure to the token to increase/decrease scale.
Modify	Extrusion	Select face, apply pressure to the extrusion token to extend the geometry.

Additional operations such as join, cut, mirror, delete, reset position, undo/redo, and save are supported through dedicated tangible tokens.

2 CONCLUSION AND FUTURE WORK

This work presents a tangible mixed reality CAD system for 3D modeling. From the participatory design workshop (n=6) and early user study (n=7), we found that a tangible token-based system requires a longer initial learning period compared to the controller-based interface. Participants reported that the tangible token-based interface was preferred by beginners over controller-based or traditional CAD software. While tangible interaction was preferred for early and mid-level modeling tasks, controller-based interaction was favored for complex operations requiring higher precision.

In future work, we plan to expand BoxCraft with additional modeling operations and further refine the tangible interaction design to improve learnability and efficiency. ArUco markers were used for reliable early prototyping, and we will explore more intuitive tangible designs to reduce cognitive load.

REFERENCES

- [1] S. Lee-Remond, S. Sagot, and E. Ostrosi, "A New Design Framework for Comprehensible Graphical User Interfaces for Parametric Computer-Aided Design Tools," *Comput Aided Des Appl*, pp. 150–179, Aug. 2024, doi: 10.14733/cadaps.2025.150-179.
- [2] T. T. H. Tran, G. Foucault, and R. Pinqu  , "Benchmarking of 3D Modelling in Virtual Reality," *Comput Aided Des Appl*, vol. 19, no. 6, pp. 1184–1190, 2022, doi: 10.14733/cadaps.2022.1184-1190.